Fractions, Decimals and Percentages

Name _____ **Date** _____ No calculator Version 1

KEY WORDS fraction · decimal · percentage · per cent · convert · equivalent

A Fractions and decimals ●○○ ANSWER ON THE SHEET

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. To go back, use place value: $0.6 = \frac{6}{10} = \frac{3}{5}$.

1. Write each fraction as a decimal.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{2}{5}$ = _____ **d)** $\frac{7}{10}$ = _____

e) $\frac{1}{8}$ = _____ **f)** $\frac{9}{20}$ = _____ **g)** $\frac{3}{25}$ = _____ **h)** $\frac{5}{8}$ = _____

2. Write each decimal as a fraction in its simplest form.

a) 0.3 = _____ **b)** 0.5 = _____ **c)** 0.25 = _____ **d)** 0.8 = _____

e) 0.45 = _____ **f)** 0.06 = _____ **g)** 0.125 = _____ **h)** 0.72 = _____

B Decimals and percentages ●○○ ANSWER ON THE SHEET

Remember: Per cent means "out of 100". Multiply a decimal by 100 to get a percentage, and divide a percentage by 100 to get a decimal.

3. Write each decimal as a percentage.

a) 0.45 = _____ **b)** 0.7 = _____ **c)** 0.08 = _____ **d)** 0.125 = _____

e) 1.5 = _____ **f)** 0.006 = _____ **g)** 0.99 = _____ **h)** 2.4 = _____

4. Write each percentage as a decimal.

a) 25% = _____ **b)** 60% = _____ **c)** 7% = _____ **d)** 12.5% = _____

e) 130% = _____ **f)** 0.5% = _____ **g)** 99% = _____ **h)** 240% = _____

C Fractions and percentages ●●○ ANSWER ON THE SHEET

5. Write each fraction as a percentage.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{2}{5}$ = _____ **d)** $\frac{3}{10}$ = _____

e) $\frac{7}{20}$ = _____ **f)** $\frac{9}{25}$ = _____ **g)** $\frac{3}{8}$ = _____ **h)** $\frac{7}{50}$ = _____

6. Write each percentage as a fraction in its simplest form.

a) 50% = _____ **b)** 25% = _____ **c)** 40% = _____ **d)** 90% = _____

e) 15% = _____ **f)** 64% = _____ **g)** 5% = _____ **h)** 12.5% = _____

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D Putting them in order ●●○ — WORK IN YOUR BOOK

> **Remember:** Change every number to the same form (decimals are usually easiest), then compare.

7. Write each list in order, smallest first.

a) $0.4, \quad 35\%, \quad \frac{3}{8}$

b) $\frac{3}{5}, \quad 0.65, \quad 62\%$

c) $7\%, \quad 0.7, \quad \frac{7}{20}$

d) $\frac{1}{3}, \quad 0.3, \quad 35\%$

E Percentage of an amount ●●○ — WORK IN YOUR BOOK

> **Remember:** Change the percentage to a decimal and multiply: 35% of 80 is 0.35×80.

8. Work out.

a) 50% of 84

b) 10% of 230

c) 25% of 48

d) 20% of 65

e) 75% of 120

f) 5% of 360

g) 30% of \$250

h) 15% of \$80

i) 12% of 400 g

j) 35% of 60 kg

k) 8% of 150

l) 120% of 45

F One amount as a percentage of another ●●● — WORK IN YOUR BOOK

> **Remember:** Write the two amounts as a fraction, then multiply by 100.

9. Answer each question.

a) What percentage of 50 is 12?

b) What percentage of 40 is 18?

c) What percentage of 250 is 45?

d) What percentage of 20 is 7?

e) Kiri scored 27 out of 60 in a test. What percentage is that?

f) A shop sold 36 of its 150 pies before noon. What percentage is that?

★ Extension — FOR EARLY FINISHERS

10. Mere scored 32 out of 40 in a maths test and 27 out of 36 in a science test. In which test did she do better? Use percentages.

11. 30% of a number is 45. Find the number.

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Fractions, Decimals and Percentages

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS fraction · decimal · percentage · per cent · convert · equivalent

A Fractions and decimals ●○○ — ANSWER ON THE SHEET

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. To go back, use place value: $0.6 = \frac{6}{10} = \frac{3}{5}$.

1. Write each fraction as a decimal.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{3}{5}$ = _____ **d)** $\frac{9}{10}$ = _____

e) $\frac{3}{8}$ = _____ **f)** $\frac{7}{20}$ = _____ **g)** $\frac{6}{25}$ = _____ **h)** $\frac{7}{8}$ = _____

2. Write each decimal as a fraction in its simplest form.

a) 0.7 = _____ **b)** 0.2 = _____ **c)** 0.75 = _____ **d)** 0.4 = _____

e) 0.35 = _____ **f)** 0.08 = _____ **g)** 0.375 = _____ **h)** 0.64 = _____

B Decimals and percentages ●○○ — ANSWER ON THE SHEET

Remember: Per cent means "out of 100". Multiply a decimal by 100 to get a percentage, and divide a percentage by 100 to get a decimal.

3. Write each decimal as a percentage.

a) 0.35 = _____ **b)** 0.9 = _____ **c)** 0.03 = _____ **d)** 0.375 = _____

e) 1.2 = _____ **f)** 0.004 = _____ **g)** 0.81 = _____ **h)** 3.5 = _____

4. Write each percentage as a decimal.

a) 45% = _____ **b)** 30% = _____ **c)** 9% = _____ **d)** 37.5% = _____

e) 160% = _____ **f)** 0.2% = _____ **g)** 88% = _____ **h)** 310% = _____

C Fractions and percentages ●●○ — ANSWER ON THE SHEET

5. Write each fraction as a percentage.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{4}{5}$ = _____ **d)** $\frac{7}{10}$ = _____

e) $\frac{11}{20}$ = _____ **f)** $\frac{8}{25}$ = _____ **g)** $\frac{5}{8}$ = _____ **h)** $\frac{9}{50}$ = _____

6. Write each percentage as a fraction in its simplest form.

a) 75% = _____ **b)** 20% = _____ **c)** 60% = _____ **d)** 30% = _____

e) 35% = _____ **f)** 48% = _____ **g)** 4% = _____ **h)** 37.5% = _____

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D Putting them in order ●●○

WORK IN YOUR BOOK

> **Remember:** Change every number to the same form (decimals are usually easiest), then compare.

7. Write each list in order, smallest first.

a) $0.6, \quad 65\%, \quad \frac{5}{8}$

b) $\frac{2}{5}, \quad 0.45, \quad 42\%$

c) $9\%, \quad 0.9, \quad \frac{9}{20}$

d) $\frac{2}{3}, \quad 0.6, \quad 65\%$

E Percentage of an amount ●●○

WORK IN YOUR BOOK

> **Remember:** Change the percentage to a decimal and multiply: 35% of 80 is 0.35×80.

8. Work out.

a) 50% of 96

b) 10% of 470

c) 25% of 64

d) 20% of 85

e) 75% of 160

f) 5% of 240

g) 30% of \$350

h) 15% of \$60

i) 12% of 300 g

j) 45% of 80 kg

k) 6% of 250

l) 130% of 40

F One amount as a percentage of another ●●●

WORK IN YOUR BOOK

> **Remember:** Write the two amounts as a fraction, then multiply by 100.

9. Answer each question.

a) What percentage of 50 is 16?

b) What percentage of 40 is 14?

c) What percentage of 250 is 35?

d) What percentage of 20 is 9?

e) Kiri scored 33 out of 60 in a test. What percentage is that?

f) A shop sold 48 of its 150 pies before noon. What percentage is that?

★ Extension

FOR EARLY FINISHERS

10. Mere scored 18 out of 24 in a maths test and 28 out of 35 in a science test. In which test did she do better? Use percentages.

11. 30% of a number is 36. Find the number.

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